EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 2.01

C Algebraic Fractions, Quadratics & Simultaneous Systems

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

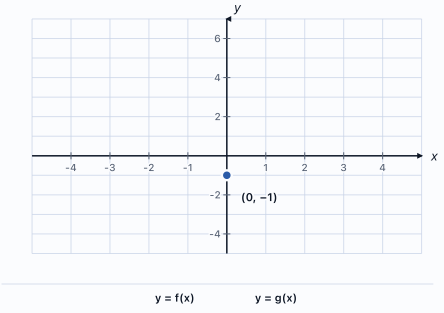
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 B	Q2 C	Q3 D	Q4 A	Q5 B
Q6 C	Q7 D	Q8 A	Q9 A	Q10 C
Q11 A	Q12 B	Q13 D	Q14 C	Q15 C
Q16 B	Q17 D	Q18 B	Q19 B	Q20 B
Q21 D	Q22 C	Q23 B	Q24 B	Q25 B

Algebraic Fractions, Quadratics & Simultaneous Systems Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 8x + 28$ in the domain of Algebraic Fractions, Quadratics & Simultaneous Systems. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(6)$.



A 32

B 28 ✓

C 16

D 26

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Complete the square: $x^2 - 8x + 28 = (x-4)^2 - 16 + 28 = (x-4)^2 + 12$, giving minimum value $k = 12$. **[M1]**
2. Evaluate at $x = 6$: $f_1(6) = (2)^2 + 12 = 16$. **[M1]**
3. Sum the two quantities: $k + f_1(6) = 12 + 16 = 28$. **[A1]**

- **Final Verified Answer:** 28.

2 MCQ

A mathematical model in Algebraic Fractions, Quadratics & Simultaneous Systems is defined by the cubic function $y_2(x) = 3x^3 - 4x^2 + 15x - 9$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96

B 57

C 72 ✓

D 34

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 8x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 9(9) - 8(3) + 15 = 72$. **[M1][A1]**

- **Final Verified Answer:** 72.

3 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, two displacement vectors are given by $\mathbf{u}_3 = \langle 4, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A -20

B 80

C 172

D 60 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (4)(9) + (-8)(7) = 36 + (-56) = -20$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 4^2 + (-8)^2 = 80$. **[M1]**
3. Combine: $-20 + 80 = 60$. **[A1]**

- **Final Verified Answer:** 60.

4 MCQ

The linear growth profile shown in the diagram for Algebraic Fractions, Quadratics & Simultaneous Systems follows an arithmetic progression with first term $a_1 = 5$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 705 ✓

B 89

C 795

D 1335

WORKED SOLUTION & VERIFICATION

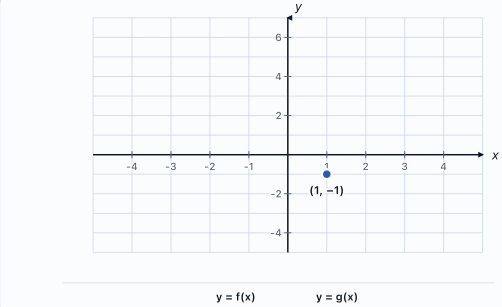
Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 5, d = 6$: $S_{15} = \frac{15}{2}[10 + 14 \times 6] = \frac{15}{2} \times 94 = 705$. **[M1][A1]**

- **Final Verified Answer:** 705.

5 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 16x)\,dx$.



A 104

B 80 ✓

C 96

D 41

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Integrate term by term: $\int (18x^2 + 16x)\,dx = [6x^3 + 8x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 8(2)^2) - 0 = 48 + 32 = 80$. **[M1][A1]**

- **Final Verified Answer:** 80.

6 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, consider the quadratic function $f_6(x) = x^2 - 10x + 109$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(12)$.

A 222

B 133

C 217 ✓

D 210

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Complete the square: $x^2 - 10x + 109 = (x-5)^2 - 25 + 109 = (x-5)^2 + 84$, giving minimum value $k = 84$. **[M1]**
2. Evaluate at $x = 12$: $f_6(12) = (7)^2 + 84 = 133$. **[M1]**
3. Sum the two quantities: $k + f_6(12) = 84 + 133 = 217$. **[A1]**

- **Final Verified Answer:** 217.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 5x^2 + 33x - 16$ studied in Algebraic Fractions, Quadratics & Simultaneous Systems. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 609

B 496

C 95

D 529 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 10x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 9(64) - 10(8) + 33 = 529$. **[M1][A1]**

- **Final Verified Answer:** 529.

8 MCQ

In Algebraic Fractions, Quadratics & Simultaneous Systems, two displacement vectors are given by $\mathbf{u}_8 = \langle 5, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 79 ✓

B 5

C 74

D 149

WORKED SOLUTION & VERIFICATION

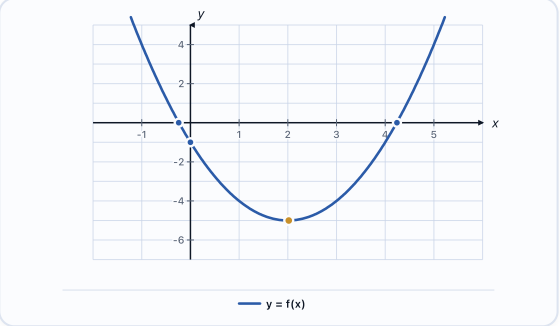
Mark Scheme & Examiner Solution (Algebraic Fractions, Quadratics & Simultaneous Systems):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (5)(8) + (-7)(5) = 40 + (-35) = 5$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 5^2 + (-7)^2 = 74$. **[M1]**
3. Combine: $5 + 74 = 79$. **[A1]**

- **Final Verified Answer:** 79.

9 MCQ

By completing the square for $y = x^2 - 4x - 1$ (Case #1), find the coordinates of the turning point of the curve.



A $(2, -5)$ ✓

B $(2, -1)$

C $(-2, 5)$

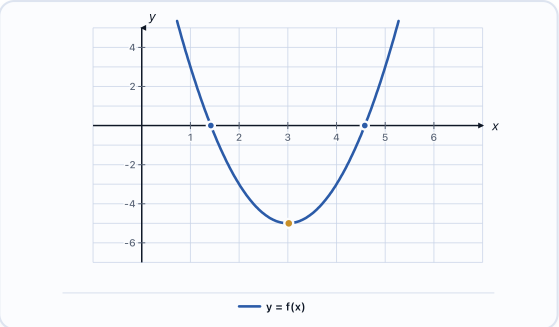
D $(-2, -5)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = (x - 2)^2 - 5$, which attains its minimum at $x = 2$ with $y = -5$, giving turning point $(2, -5)$.

10 MCQ

By completing the square for $y = 2x^2 - 12x + 13$ (Case #2), find the coordinates of the turning point of the curve.



A $(-3, 5)$

B $(3, 13)$

C $(3, -5)$ ✓

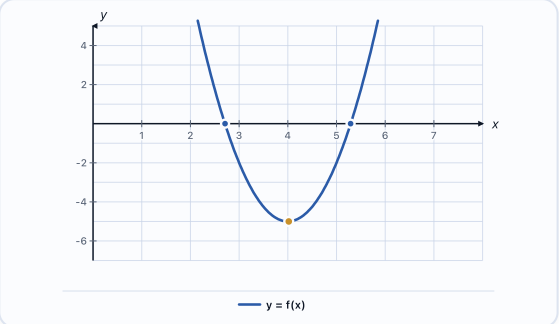
D $(-3, -5)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 2(x - 3)^2 - 5$, which attains its minimum at $x = 3$ with $y = -5$, giving turning point $(3, -5)$.

11 MCQ

By completing the square for $y = 3x^2 - 24x + 43$ (Case #3), find the coordinates of the turning point of the curve.



A $(4, -5)$ ✓

B $(-4, -5)$

C $(-4, 5)$

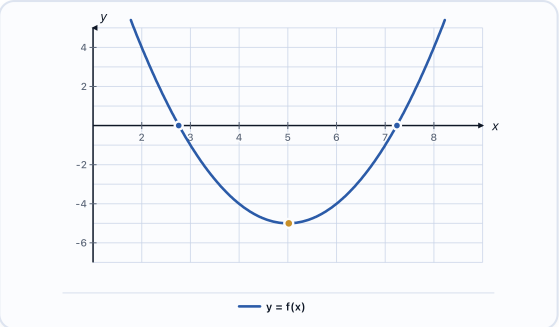
D $(4, 43)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 3(x - 4)^2 - 5$, which attains its minimum at $x = 4$ with $y = -5$, giving turning point $(4, -5)$.

12 MCQ

By completing the square for $y = x^2 - 10x + 20$ (Case #4), find the coordinates of the turning point of the curve.



A $(-5, 5)$

B $(5, -5)$ ✓

C $(-5, -5)$

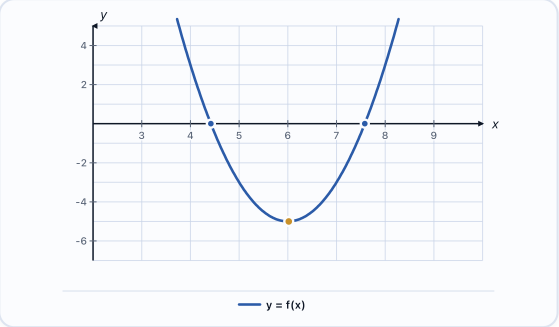
D $(5, 20)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = (x - 5)^2 - 5$, which attains its minimum at $x = 5$ with $y = -5$, giving turning point $(5, -5)$.

13 MCQ

By completing the square for $y = 2x^2 - 24x + 67$ (Case #5), find the coordinates of the turning point of the curve.



A $(-6, -5)$

B $(-6, 5)$

C $(6, 67)$

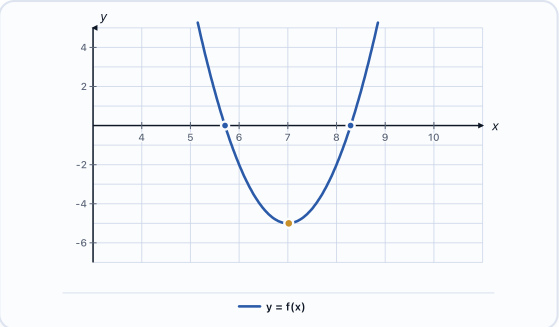
D $(6, -5)$ ✓

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 2(x - 6)^2 - 5$, which attains its minimum at $x = 6$ with $y = -5$, giving turning point $(6, -5)$.

14 MCQ

By completing the square for $y = 3x^2 - 42x + 142$ (Case #6), find the coordinates of the turning point of the curve.



A $(-7, -5)$

B $(-7, 5)$

C $(7, -5)$ ✓

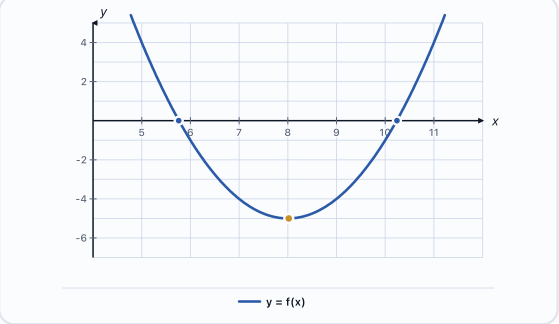
D $(7, 142)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 3(x - 7)^2 - 5$, which attains its minimum at $x = 7$ with $y = -5$, giving turning point $(7, -5)$.

15 MCQ

By completing the square for $y = x^2 - 16x + 59$ (Case #7), find the coordinates of the turning point of the curve.



A $(-8, 5)$

B $(8, 59)$

C $(8, -5)$ ✓

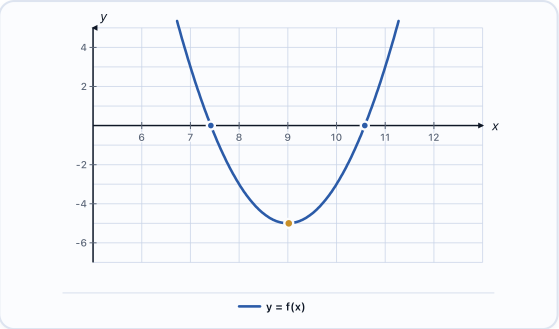
D $(-8, -5)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = (x - 8)^2 - 5$, which attains its minimum at $x = 8$ with $y = -5$, giving turning point $(8, -5)$.

16 MCQ

By completing the square for $y = 2x^2 - 36x + 157$ (Case #8), find the coordinates of the turning point of the curve.



A $(9, 157)$

B $(9, -5)$ ✓

C $(-9, -5)$

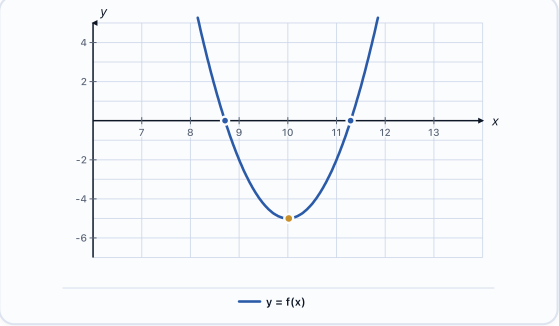
D $(-9, 5)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 2(x - 9)^2 - 5$, which attains its minimum at $x = 9$ with $y = -5$, giving turning point $(9, -5)$.

17 MCQ

By completing the square for $y = 3x^2 - 60x + 295$ (Case #9), find the coordinates of the turning point of the curve.



A $(-10, -5)$

B $(10, 295)$

C $(-10, 5)$

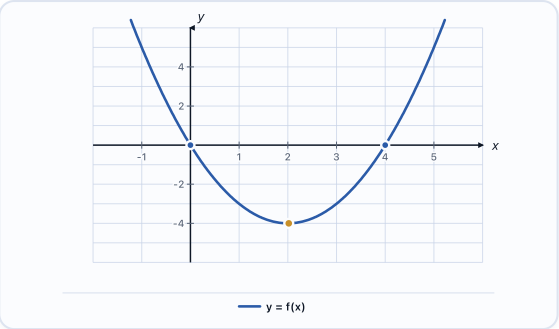
D $(10, -5)$ ✓

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 3(x - 10)^2 - 5$, which attains its minimum at $x = 10$ with $y = -5$, giving turning point $(10, -5)$.

18 MCQ

By completing the square for $y = x^2 - 4x + 0$ (Case #10), find the coordinates of the turning point of the curve.



A $(-2, -4)$

B $(2, -4)$ ✓

C $(2, 0)$

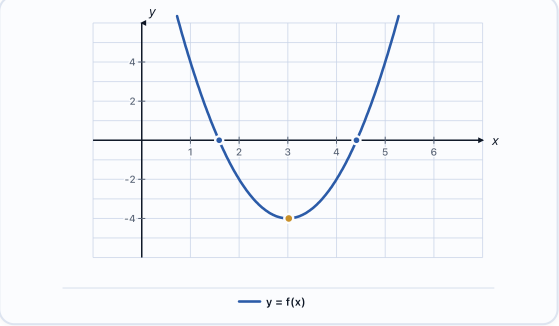
D $(-2, 4)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = (x - 2)^2 - 4$, which attains its minimum at $x = 2$ with $y = -4$, giving turning point $(2, -4)$.

19 MCQ

By completing the square for $y = 2x^2 - 12x + 14$ (Case #11), find the coordinates of the turning point of the curve.



A $(-3, 4)$

B $(3, -4)$ ✓

C $(3, 14)$

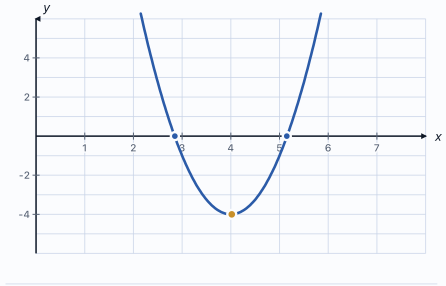
D $(-3, -4)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 2(x - 3)^2 - 4$, which attains its minimum at $x = 3$ with $y = -4$, giving turning point $(3, -4)$.

20 MCQ

By completing the square for $y = 3x^2 - 24x + 44$ (Case #12), find the coordinates of the turning point of the curve.



A $(-4, -4)$

B $(4, -4)$ ✓

C $(4, 44)$

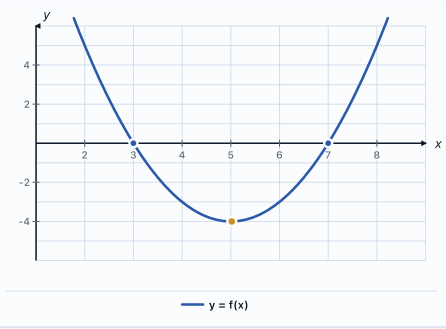
D $(-4, 4)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 3(x-4)^2 - 4$, which attains its minimum at $x = 4$ with $y = -4$, giving turning point $(4, -4)$.

21 MCQ

By completing the square for $y = x^2 - 10x + 21$ (Case #13), find the coordinates of the turning point of the curve.



A $(-5, -4)$

B $(5, 21)$

C $(-5, 4)$

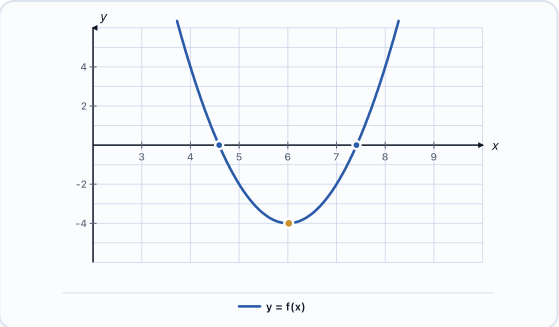
D $(5, -4)$ ✓

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = (x-5)^2 - 4$, which attains its minimum at $x = 5$ with $y = -4$, giving turning point $(5, -4)$.

22 MCQ

By completing the square for $y = 2x^2 - 24x + 68$ (Case #14), find the coordinates of the turning point of the curve.



A $(-6, 4)$

B $(6, 68)$

C $(6, -4)$ ✓

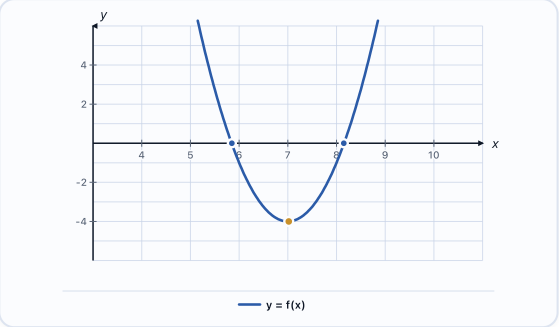
D $(-6, -4)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 2(x - 6)^2 - 4$, which attains its minimum at $x = 6$ with $y = -4$, giving turning point $(6, -4)$.

23 MCQ

By completing the square for $y = 3x^2 - 42x + 143$ (Case #15), find the coordinates of the turning point of the curve.



A $(7, 143)$

B $(7, -4)$ ✓

C $(-7, 4)$

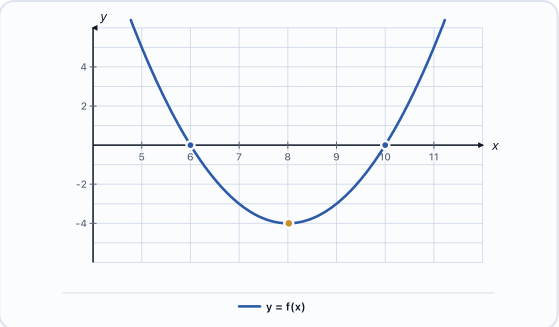
D $(-7, -4)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 3(x - 7)^2 - 4$, which attains its minimum at $x = 7$ with $y = -4$, giving turning point $(7, -4)$.

24 MCQ

By completing the square for $y = x^2 - 16x + 60$ (Case #16), find the coordinates of the turning point of the curve.



A $(-8, 4)$

B $(8, -4)$ ✓

C $(8, 60)$

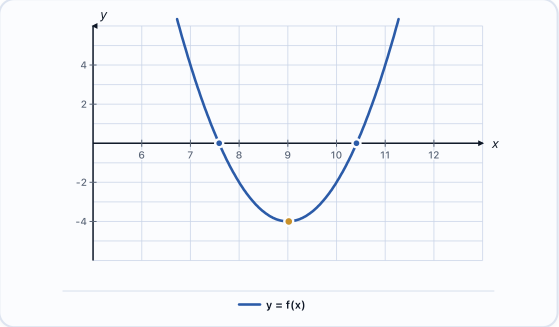
D $(-8, -4)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = (x - 8)^2 - 4$, which attains its minimum at $x = 8$ with $y = -4$, giving turning point $(8, -4)$.

25 MCQ

By completing the square for $y = 2x^2 - 36x + 158$ (Case #17), find the coordinates of the turning point of the curve.



A $(-9, 4)$

B $(9, -4)$ ✓

C $(9, 158)$

D $(-9, -4)$

WORKED SOLUTION & VERIFICATION

Completing the square gives $y = 2(x - 9)^2 - 4$, which attains its minimum at $x = 9$ with $y = -4$, giving turning point $(9, -4)$.